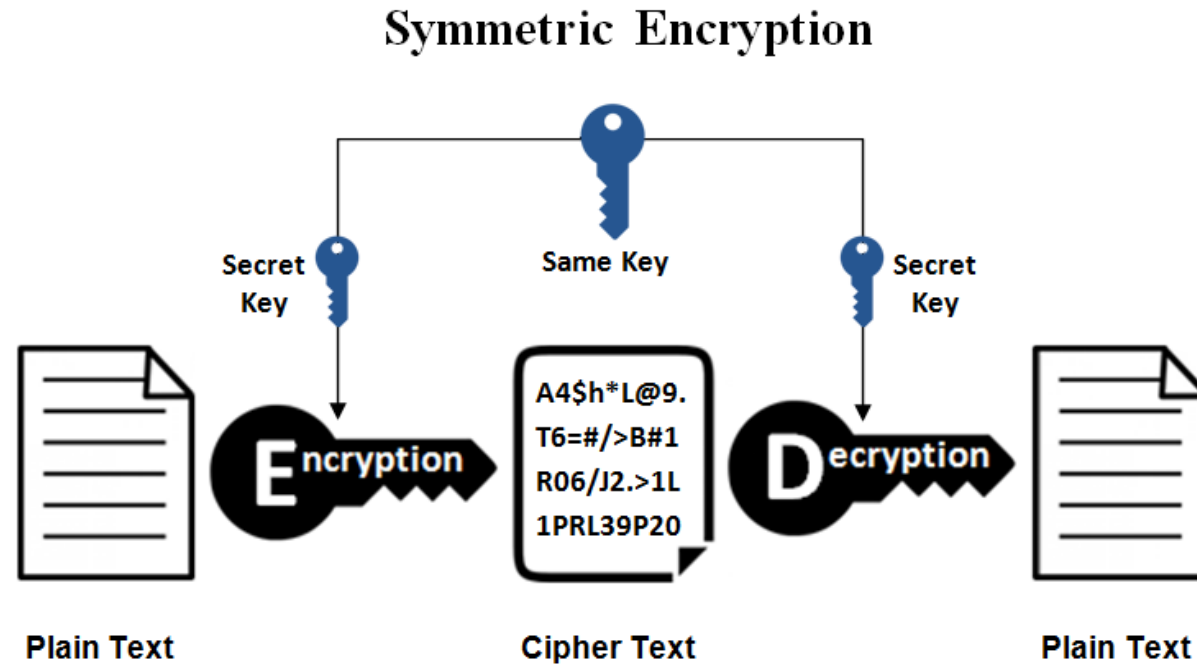


Final exam time and location

- December 17th, 2025, Wed
- 1:30-3:30 pm, 1015 ECEB
- Next class: Tianyi Liu
- No office hour this week

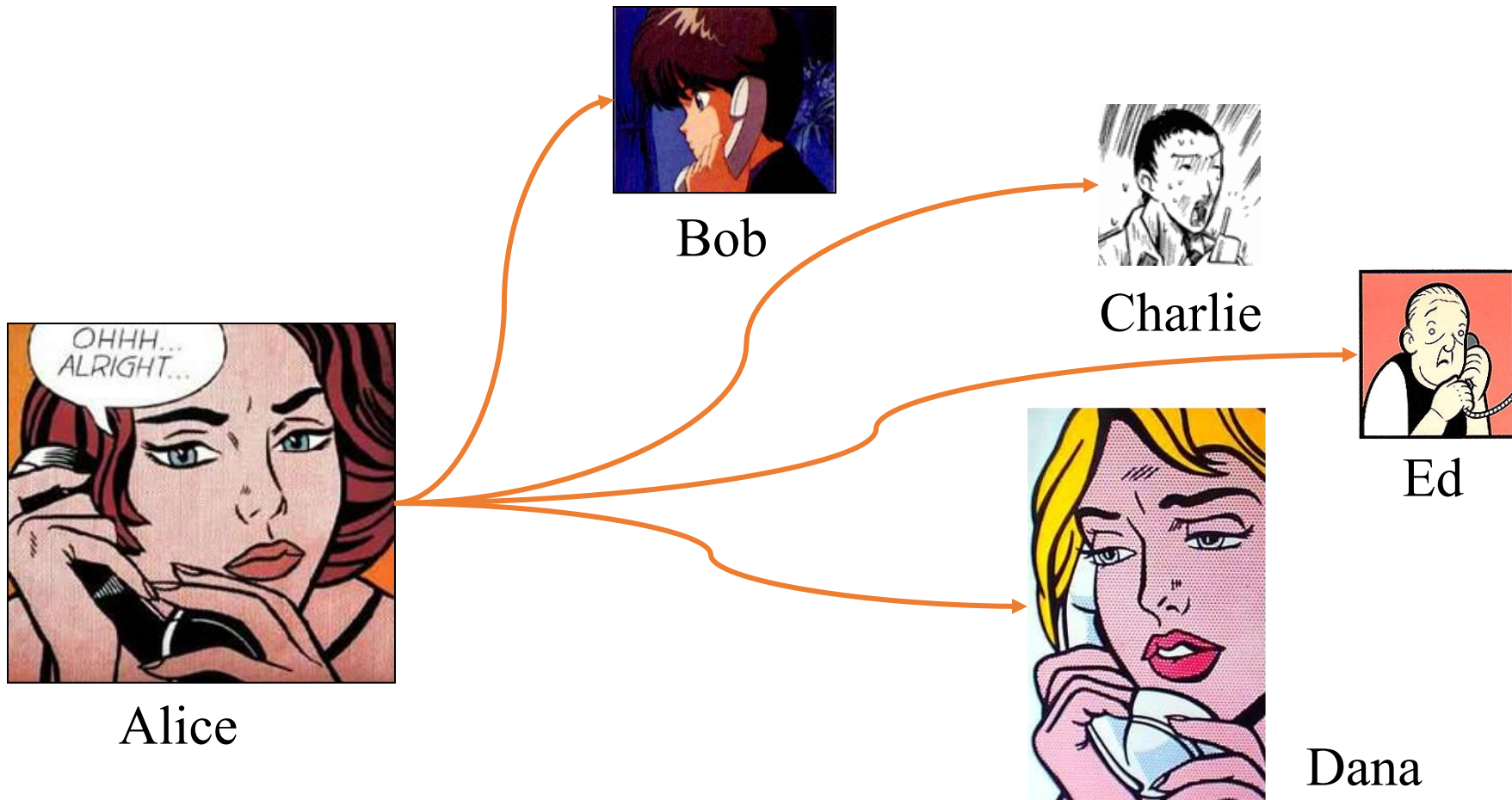
Public key cryptography

Difficulties with symmetric keys

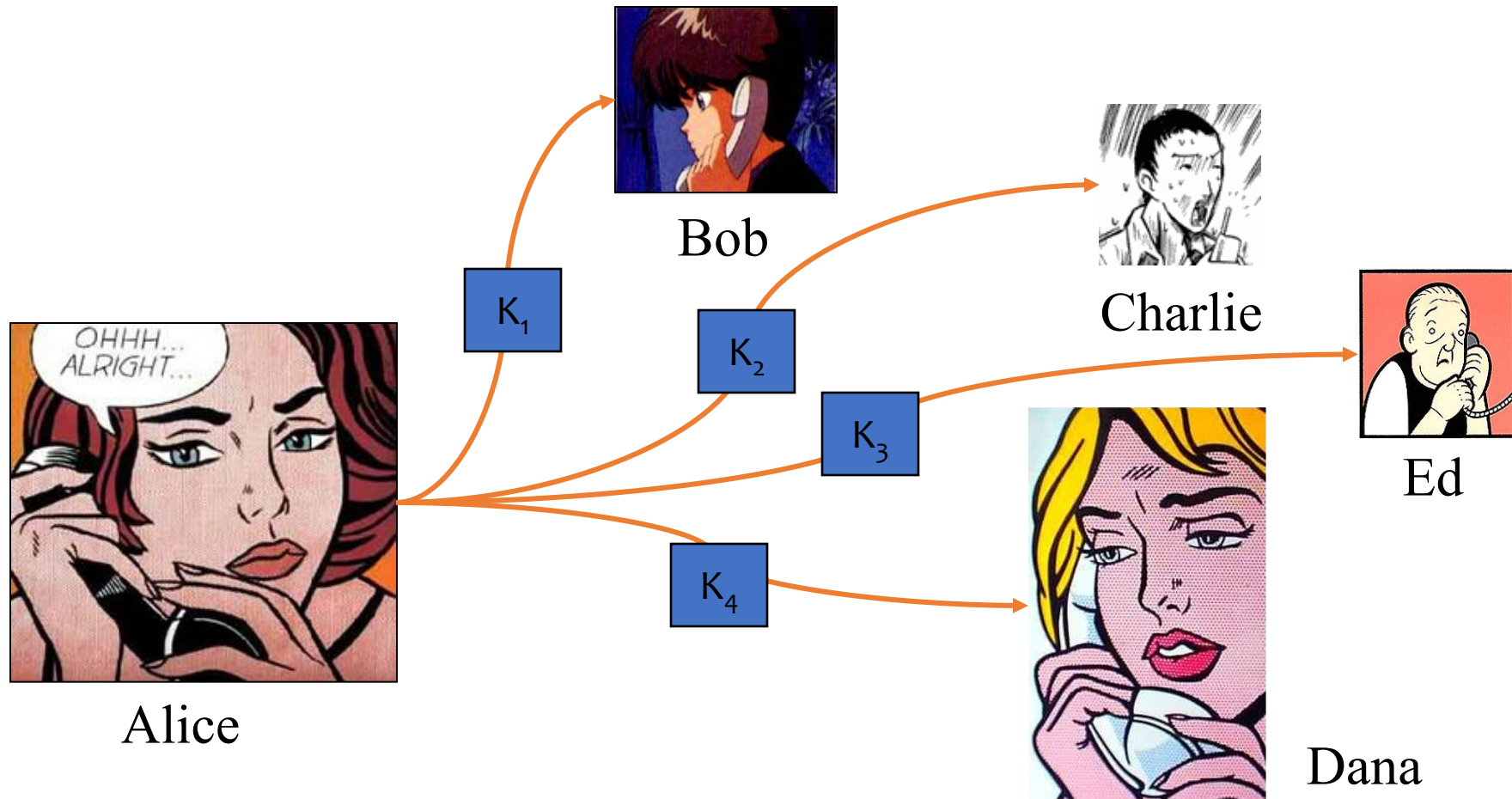


How can Alice and Bob share the secret key?

More difficulties with sym. keys



More difficulties with sym. keys



More difficulties with sym. keys



Alice

For n participants in the system, you need $O(n^2)$ keys if everyone can talk to everyone else!



Ed



Dana

Even worse for group chat



Alice



Bob



Charlie



Ed



Dana

$$\binom{n}{2} + \binom{n}{3} + \binom{n}{4} + \dots \approx 2^n$$

Solutions

- **Key Exchange**
- **Public key encryption**

Prime number

- Prime number p :
 - p is an integer
 - $p \geq 2$
 - The only divisors of p are 1 and p
- Examples
 - 2, 7, 19 are primes
 - -3, 0, 1, 6 are not primes

Prime decomposition

- Prime decomposition of a positive integer n :

$$n = p_1^{e_1} \times \dots \times p_k^{e_k}$$

- Example:

- $200 = 2^3 \times 5^2$

Modular Arithmetic

- Modulo operator for a positive integer n

$$r = a \bmod n$$

equivalent to

$$a = r + kn$$

Example:

$$29 \bmod 13 = 3$$

$$13 \bmod 13 = 0$$

$$-1 \bmod 13 = 12$$

Notation

- N denotes a positive integer
- p denotes a prime
- $\mathbb{Z}$: set of integers
- $\mathbb{Z}_N = \{0, 1, 2, \dots, N - 1\}$
- Arithmetic in $\mathbb{Z}_N$ works as you expect,
 - $x + y = y + x \pmod N$
 - $x + y + z = x + (y + z) \pmod N$
 - $x \cdot (y + z) = x \cdot y + x \cdot z \pmod N$
- But not division!
- Example $3 \cdot 2 = 6 = 15 \cdot 2 \pmod{24}$, but $3 \not\equiv 15 \pmod{24}$

Simplifying modular arithmetic

- $a + b + c \bmod N = (a + b) \bmod N + c \bmod N$

- $a \times b \times c \bmod N = (a \times b) \bmod N \times c \bmod N$

- $a^b \bmod N = \underbrace{a \times a \times \cdots \times a}_{b \text{ times}} \bmod N$

- $2^{122} \bmod 7 = ?$

Group

Group: under 1 operation $\circ$

1. **Closure**: For all a, b in G , the result of the operation, $a \circ b$, is also in G
2. **Associativity**: For all a, b and c in G , $(a \circ b) \circ c = a \circ (b \circ c)$
3. **Identity element**: There exists an element e in G such that, for every element a in G , the equation $e \circ a = a \circ e = a$ holds. Such an element is unique
4. **Inverse element**: For each a in G , there exists an element b in G , such that $a \circ b = b \circ a = e$, where e is the identity element

Examples: integer with $+$

$$\mathbb{Z}_N = \{0, 1, 2, \dots, N - 1\} \text{ with } + \text{ mod } N$$

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Commutativity: If for all a, b in G , $a \circ b = b \circ a$, the group is called **abelian**

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What about integer with $\times$?

What about $\mathbb{Z}_N = \{0, 1, 2, \dots, N - 1\}$ with $\times \bmod N$

Greatest common divisor

Def: For integers x, y : **gcd(x, y)** is the greatest common divisor of x, y

Example: $\text{gcd}(12, 18) = 6$

Computing GCD: Euclidean algorithm

Greatest common divisor

Def: For integers x, y : **gcd(x, y)** is the greatest common divisor of x, y

Example: $\text{gcd}(12, 18) = 6$

Fact: for all integers x, y there exist integers a, b such that

$$a \cdot x + b \cdot y = \text{gcd}(x, y)$$

a, b can be found efficiently using the extended Euclidean algorithm

If $\text{gcd}(x, y) = 1$ we say that x and y are **relatively prime**

Modular inversion (multiplicative)

Over the rational numbers, inverse of 2 is $1/2$. What about $\mathbb{Z}_N$?

Def: The **inverse** of x in $\mathbb{Z}_N$ is an element $y \in \mathbb{Z}_N$ s.t.

$$x \cdot y = 1 \text{ in } \mathbb{Z}_N. \quad y \text{ is denoted as } x^{-1}$$

Example: let N be an odd integer. The inverse of 2 in $\mathbb{Z}_N$ is $\frac{N+1}{2}$

Modular inversion (multiplicative)

Which elements have an inverse in $\mathbb{Z}_N$? $2 \times ??? = 1 \pmod{100}$?

Lemma: $x \in \mathbb{Z}_N$ has an inverse if and only if $\gcd(x, N) = 1$

Proof:

$$\gcd(x, N) = 1 \Rightarrow \exists a, b: a \cdot x + b \cdot N = 1 \Rightarrow a \cdot x = 1 \pmod{N}$$

$$\gcd(x, N) > 1 \Rightarrow \forall a: \gcd(a \cdot x, N) > 1 \Rightarrow a \cdot x \neq 1 \pmod{N}$$

For $x \in \mathbb{Z}_N$, can find x^{-1} using extended Euclidean algorithm.

More notation

Def: $\mathbb{Z}_N^*$ = (set of invertible elements in $\mathbb{Z}_N$)
 = $\{ x \in \mathbb{Z}_N : \gcd(x, N) = 1 \}$

Examples:

1. for prime $p = 11$, $\mathbb{Z}_p^* = \mathbb{Z}_p \setminus \{0\} = \{1, 2, \dots, 10\}$
2. $\mathbb{Z}_{12}^* = \{1, 5, 7, 11\}$

Multiplicative group $\mathbb{Z}_N^*$

- $\mathbb{Z}_N^*$ is an abelian group under multiplication mod N (for $N > 1$).

Group

Group: under 1 operation $\circ$

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Cyclic group and generator

Definition: $\mathbb{Z}_p^*$ is a cyclic group, if

$$\exists g \in \mathbb{Z}_p^* \text{ such that } \{g^0, g, g^2, \dots, g^{p-2}\} = \mathbb{Z}_p^*$$

Here $g^0 = e$

Example $p = 7$: $\{1, 3, 3^2, 3^3, 3^4, 3^5\} = \{1, 3, 2, 6, 4, 5\} = \mathbb{Z}_7^*$

Not every element is a generator: $\{1, 2, 2^2, 2^3, 2^4, 2^5\} = \{1, 2, 4\}$

g is called a **generator** of $\mathbb{Z}_p^*$

How many generators $\mathbb{Z}_p^*$ has ?

Cyclic group and generator

- Also defined for +
- E.g., $\mathbb{Z}_7 = \{0,1,2,3,4,5,6\}$ under +
- 1 is a generator, 2 is also a generator

Cyclic group and generator

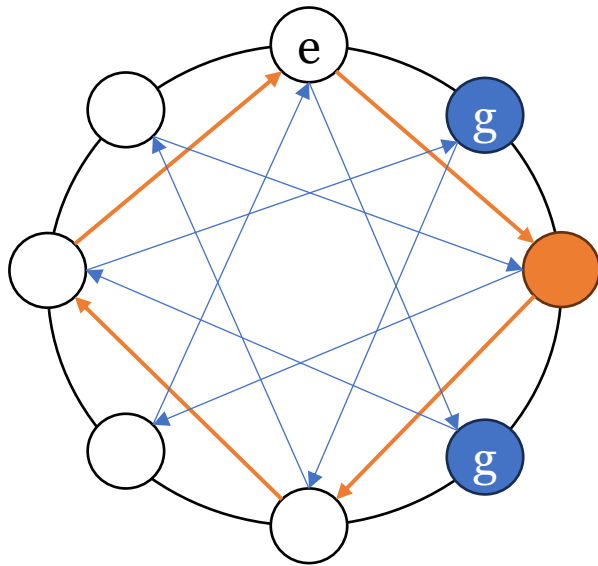
Defined for any group

Definition: $\mathbb{G}$ is a cyclic group, if

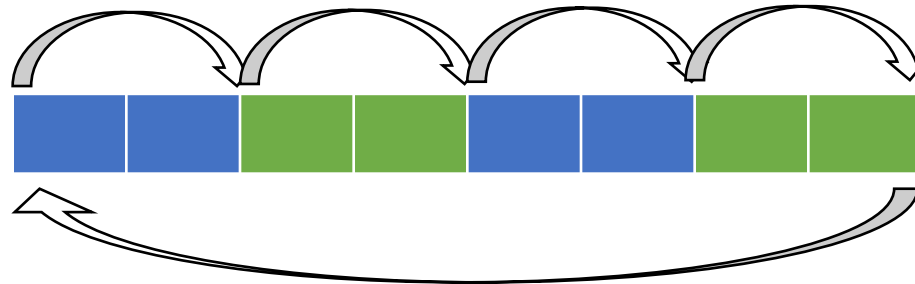
$\exists g \in \mathbb{G}$ such that $\{e, g, g^2, \dots, g^{|\mathbb{G}|-1}\} = \mathbb{G}$ by applying the group operation on g

For any $d < |\mathbb{G}| - 1$, $g^d \neq e$

Who could be a generator?



- When $N = 8$
 - $N = 8$ can be written as $N = 2 \times 4$.
 - $n = 2$ can be written as $n = 2 \times 1$



- All have **stride = 2** and never touch the points inside the stride!
- What's the condition of being a generator?

Order of a group and an element

- The **order** of a group is its number of elements. E.g., $|\mathbb{Z}_p^*| = p - 1$
- $x^{|\mathbb{G}|} = e$ for any $x \in \mathbb{G}$
- The **order** of an element is the minimum number m such that $x^m = 1$ in $\mathbb{G}$

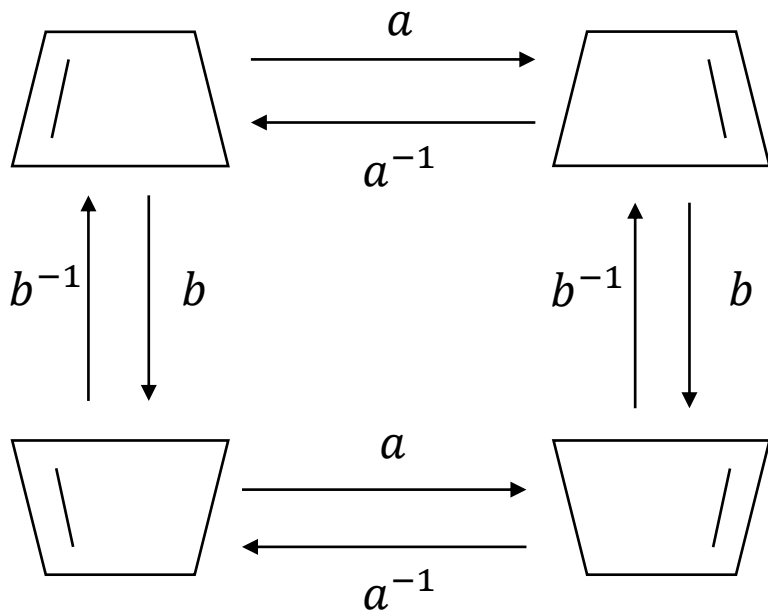
Examples: order of 3 in $\mathbb{Z}_7^*$ $\text{ord}_7(3) = 6$;

order of 2 in $\mathbb{Z}_7^*$ $\text{ord}_7(2) = 3$;

order of 1 in $\mathbb{Z}_7^*$ $\text{ord}_7(1) = 1$

Note not all groups are cyclic group!

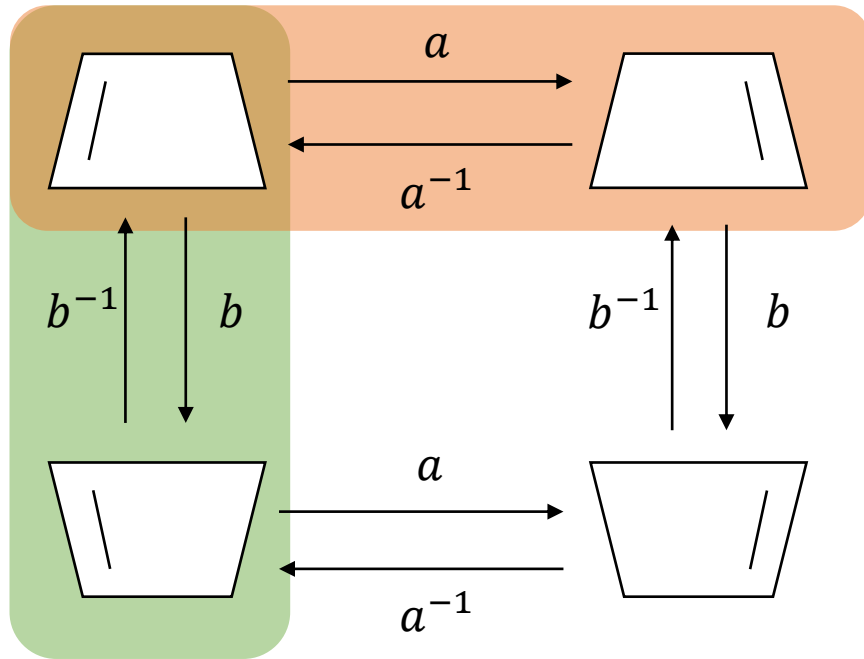
- Cyclic group can be generated by a single element.



- $a^2 = e, b^2 = e \Rightarrow a = a^{-1}, b = b^{-1}$
- The whole group (order = 4).
 $\{e, a, b, ab\}$
- $\langle a \rangle = \{e, a\}, \text{ord}(a) = 2$
- $\langle b \rangle = \{e, b\}, \text{ord}(b) = 2$

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Subgroup

- $\mathbb{H}$ is a subgroup of $\mathbb{G}$ if $\mathbb{H}$ is a subset of $\mathbb{G}$ and is also a group under the same group operation of $\mathbb{G}$
 - $e \in \mathbb{H}$
 - $a \in \mathbb{H}, a^{-1} \in \mathbb{H}$
 - $a, b \in \mathbb{H}, ab \in \mathbb{H}$
- Every element of $\mathbb{G}$ generates a subgroup
E.g., 2 in $\mathbb{Z}_7^*$: $\{1, 2, 4\}$
Why?

The order of the subgroup is the order of the element in the group.

Order of a group and an element

- **Lagrange's Theorem** : $\forall x \in \mathbb{G} : \text{ord}(x)$ divides $|\mathbb{G}|$
- Proof (only for abelian group, though the theorem is for all groups).
 - Firstly, prove $x^{|\mathbb{G}|} = e$. (We mentioned before, but we didn't prove it)
 - Secondly, since we have $x^{\text{ord}(x)} = 1$, then any $x^k = x^{k \bmod \text{ord}(x)}$.
 - The first point just means $x^N = x^{N \bmod \text{ord}(x)} = x^0 = e$!
 - Therefore $N \bmod \text{ord}(x) = 0$, which means $\text{ord}(x) \mid N$.

Fermat's Little Theorem

Theorem

Let p be a prime. For all $x \in \mathbb{Z}_p^*$, we have $x^{p-1} = 1 \bmod p$

- Example ($p = 5$):

$$1^4 \bmod 5 = 1$$

$$2^4 \bmod 5 = 16 \bmod 5 = 1$$

$$3^4 \bmod 5 = 81 \bmod 5 = 1$$

$$4^4 \bmod 5 = 256 \bmod 5 = 1$$

Discrete-logarithm

$\mathbb{Z}_p^*$ has an alternative representation as the powers of g :
 $\{1, g, g^2, g^3, \dots, g^{p-2}\}$

How Many generators $\mathbb{Z}_p^*$ has ?

Discrete-log: given $a \in \mathbb{Z}_p^*$, find x s. t. $g^x = a$

Example: what is $3^x = 6 \bmod 7$

Discrete-logarithm example

- $\mathbb{Z}_{11}^*$ with $g = 2$

x	1	2	3	4	5	6	7	8	9	10
$2^x \bmod 11$	2	4	8	5	10	9	7	3	6	1

Discrete-logarithm example

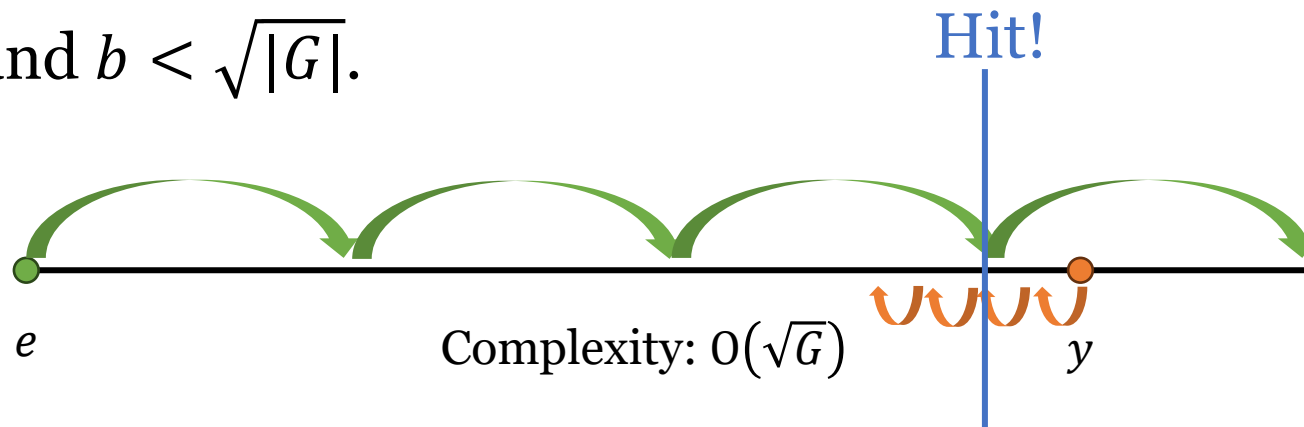
- $\mathbb{Z}_{11}^*$ with $g = 2$

y	1	2	3	4	5	6	7	8	9	10
$Dlog_2(y)$	10	1	8	2	4	9	7	3	6	5

Find x such that
 $2^x = y \bmod 11$

How to solve DL problem?

- Remember the **BSGS algorithm** in the first class?
- To solve x in $g^x = y$, it is equivalent to solve $g^{a\sqrt{|G|}+b} = y$.
 - Because all numbers in $[0, \dots, |G|)$ can be denoted by $a\sqrt{|G|} + b$ for some $a < \sqrt{|G|}$ and $b < \sqrt{|G|}$.



There are other efficient algorithms but I'm not familiar with ☹.

Discrete-log assumption:

Discrete-log is hard, exponential time to compute, brute-force

Definition: we say that DLOG is hard in a finite cyclic group $\mathbb{G}$ if for all PPT algorithm A ,

$$\Pr[A(\mathbb{G}, q, g, g^x) = x] \leq \text{negl}(n)$$

where g is a generator, q is the order of the group with $\|q\| = n$, and $x \in \mathbb{Z}_q$ is chosen uniformly